



Solve the given problems:

1. The negative root of the smallest magnitude of the equation
$$f(x) = 3x^3 + 10x^2 + 10x + 7 = 0$$
 is to be obtained.
 - (i) Find an interval of unit length which contains this root.
 - (ii) Perform two iterations of the bisection method.
 - (iii) Taking the end points of the last interval (obtained by bisection method) as initial approximations, perform three iterations of the secant method.
2. The smallest positive root of the equation
$$f(x) = x^4 - 3x^2 + x - 10 = 0$$
 is to be obtained.
 - (i) Find an interval of unit length which contains this root.
 - (ii) Perform two iterations of the bisection method.
 - (iii) Taking the mid point of the last interval (obtained by bisection method) as initial approximations, perform three iterations of the Newton method or Newton-Raphson method.
3. Define the function g by $g(0) = 0, g(x) = -x \sin^2(1/x)$ for $0 < x \leq 1$. Show that g is continuous, and that 0 is the only fixed point of g in the interval $[0,1]$. By considering the iteration $x_{n+1} = g(x_n), n = 0,1,2, \dots$, starting, first from $x_0 = 1/(k\pi)$, and then from $x_0 = 2/((2k+1)\pi)$, where k is an integer, show that the fixed point is neither stable nor unstable.
4. Find the iterative methods based on the Newton or Newton Raphson method for finding $\sqrt{N}, 1/N, N^{\frac{1}{3}}$, where N is a positive real number. Apply the methods to $N = 18$ to obtain the results correct to two decimals.

5. Newton's method is applied to the solution of $e^x - x - 2 = 0$. Show that if the starting value is positive, the iteration converges to the positive solution, and if the starting value is negative it converges to the negative solution. Discuss the convergency of the method if the starting value (i) $x_0 = 100$ and (ii) $x_0 = -100$. How many iterations would be required to obtain the solution to six decimal digits in those two cases?
6. Suppose that $f(\xi) = f'(\xi) = 0$, so that f has a double root at ξ , and that f'' is defined and continuous in a neighbourhood of ξ . If $\{x_k\}$ is a sequence obtained by Newton's method, show that
$$\xi - x_{k+1} = -\frac{\frac{1}{2}((\xi - x_k)^2 f''(\eta_k))}{f'(x_k)} = \frac{\frac{1}{2}(\xi - x_k)f''(\chi_k)}{f''(\chi_k)},$$
 where η_k and χ_k both lie between ξ and x_k . Suppose, further that $0 < m < |f''(x)| < M \quad \forall x$ in the interval $[\xi - \delta, \xi + \delta]$ for some $\delta > 0$, where $M < 2m$; show that if x_0 lies in this interval the iteration converges to ξ , and that convergence is linear, with the rate $\log_{10} 2$. Verify this conclusion by finding the solution $e^x = 1 + x$, beginning from $x_0 = 1$.
7. The equation $f(x) = 0$ where $f(x) = 0.1 - x + \frac{x^2}{(2!)^2} - \frac{x^3}{(3!)^2} + \frac{x^4}{(4!)^2} - \dots$ has one root in the interval $(0,1)$. Calculate this root correct to 5 decimals.
8. Show that the equation $f(x) = \cos\left(\frac{\pi(x+1)}{8}\right) + 0.148x - 0.9062 = 0$ has one root in the interval $(-1,0)$ and one in $(0,1)$. Calculate the negative root correct to 4 decimals.
9. (a) For the equation $\log_e(x) = x^2 - 1$, find two positive roots in the interval $(0,1)$ by using the fixed-point simple iteration method and discuss the convergency of this method for finding each root. Discuss the stability of the fixed points, whether these are stable or unstable.
- (b) Determine for which initial approximation x_0 , the iteration
$$x_{n+1} = x_n - \frac{x_n - \exp(x_n^2 - 1)}{1 - 2x_n \exp(x_n^2 - 1)}$$
 converges to the roots of above equation given in (9a).

11. Apply the Newton method to the equation $f(x) = c$, where c is real valued constant and $f(x) = \begin{cases} \cos x, & \text{when } |x| \leq 1, \\ \cos x + (x^2 - 1)^2, & \text{when } |x| \geq 1. \end{cases}$ For which c , we can get the sequence $\{x_n\}$, where $x_n = (-1)^n$, starting with $x_0 = 1$ and the calculations are carried out with no errors?
12. Determine the order of convergence of the iterative method
$$x_{k+1} = \frac{(x_0 f(x_k) - x_k f(x_0))}{f(x_k) - f(x_0)}$$
 for finding a simple root of the equation $f(x) = 0$.
13. Let $x = \xi$ be a simple root of the equation $f(x) = 0$. For the root finding iterative method $x_{k+1} = x_k - \frac{(f(x_k))^2}{f(x_k) - f(x_k - f(x_k))}$, $k = 0, 1, 2, \dots$, find the order of the convergence and compare the convergence properties with those of Newton's method.
14. The equation $f(x) = 0$ has a simple root in the interval $(1, 2)$. The function $f(x)$ is such that $|f'(x)| \geq 4$ and $|f''(x)| \leq 3$ for all x in $(1, 2)$. Assuming that the Newton's method converges for all initial approximations in $(1, 2)$, find the maximum number of iterations required to obtain the root correct to 6 decimal places after rounding.
15. We want to find the root of the equation $\exp(-x) = 3 \log_e(x)$ using the formula $x_{n+1} = x_n - \frac{(3 \log_e(x_n) - \exp(-x_n))}{p}$, $n = 0, 1, 2, \dots$. Show that $p = 3$ gives rapid convergence.
16. (a) The equation $2e^{-x} = \frac{1}{x+2} + \frac{1}{x+1}$ has two roots greater than -1 . Calculate these roots correct to five decimal places.
- (b) find the positive root of the equation $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} e^{0.3x}$ correct to six decimal places.

If want get more problems, please find some more problems in the book -

AN INTRODUCTION TO NUMERICAL ANALYSIS by Atkinson